

现场检查记录扫描件

SYNTHETIC_TEST_ONLY | 合成测试数据 | 第 1 页

S40_MITIGATION_PACKAGE × D00_CLEAN | 基准日期 2026-08-26 | 禁止用于真实工程

engineering_indicator.construction.backfill_type | 回填材料类型 | IMPORTANT

engineering_indicators.observations_by_segment.*.construction.backfill_type

值: {"by_archetype":{"DENSE":"筛分细土","GEOHAZARD":"筛分细土","INDUSTRIAL":"筛分细土","RURAL":"筛分细土","VILLAGE":"筛分细土"},"segment_overrides":{}}

engineering_indicator.construction.casing_condition | 套管状态 | IMPORTANT

engineering_indicators.observations_by_segment.*.construction.casing_condition

值: {"by_archetype":{"DENSE":"GOOD","GEOHAZARD":"GOOD","INDUSTRIAL":"GOOD","RURAL":"GOOD","VILLAGE":"GOOD"},"segment_overrides":{}}

engineering_indicator.construction.casing_count_per_km | 套管密度 | IMPORTANT

engineering_indicators.observations_by_segment.*.construction.casing_count_per_km

值: {"by_archetype":{"DENSE":0.0,"GEOHAZARD":0.0,"INDUSTRIAL":0.0,"RURAL":0.0,"VILLAGE":0.0},"segment_overrides":{}}

engineering_indicator.construction.cold_bend_radius_ratio | 冷弯管弯曲半径比 | IMPORTANT

engineering_indicators.observations_by_segment.*.construction.cold_bend_radius_ratio

值: {"by_archetype":{"DENSE":40.0,"GEOHAZARD":40.0,"INDUSTRIAL":40.0,"RURAL":40.0,"VILLAGE":40.0},"segment_overrides":{}}

engineering_indicator.construction.concrete_slab_coverage_fraction | 盖板保护覆盖率 | IMPORTANT

engineering_indicators.observations_by_segment.*.construction.concrete_slab_coverage_fraction

值: {"by_archetype":{"DENSE":0.0,"GEOHAZARD":0.0,"INDUSTRIAL":0.0,"RURAL":0.0,"VILLAGE":0.0},"segment_overrides":{}}

engineering_indicator.construction.construction_contractor | 施工单位 | IMPORTANT

engineering_indicators.observations_by_segment.*.construction.construction_contractor

值: {"by_archetype":{"DENSE":"SYNTHETIC_CONSTRUCTION_CO","GEOHAZARD":"SYNTHETIC_CONSTRUCTION_CO","INDUSTRIAL":"SYNTHETIC_CONSTRUCTION_CO","RURAL":"SYNTHETIC_CONSTRUCTION_CO","VILLAGE":"SYNTHETIC_CONSTRUCTION_CO"},"segment_overrides":{}}

engineering_indicator.construction.construction_nonconformity_count | 施工不符合项数量 | IMPORTANT

engineering_indicators.observations_by_segment.*.construction.construction_nonconformity_count

值: {"by_archetype":{"DENSE":0,"GEOHAZARD":0,"INDUSTRIAL":0,"RURAL":0,"VILLAGE":0},"segment_overrides":{}}

engineering_indicator.construction.field_bend_count_per_km | 现场弯管密度 | IMPORTANT

engineering_indicators.observations_by_segment.*.construction.field_bend_count_per_km

值: {"by_archetype":{"DENSE":0.0,"GEOHAZARD":0.0,"INDUSTRIAL":0.0,"RURAL":0.0,"VILLAGE":0.0},"segment_overrides":{}}

现场检查记录扫描件

SYNTHETIC_TEST_ONLY | 合成测试数据 | 第 2 页

S40_MITIGATION_PACKAGE × D00_CLEAN | 基准日期 2026-08-26 | 禁止用于真实工程

engineering_indicator.construction.girth_weld_count_per_km | 环焊缝密度 | IMPORTANT

engineering_indicators.observations_by_segment.*.construction.girth_weld_count_per_km

值: {"by_archetype":{"DENSE":0.0,"GEOHAZARD":0.0,"INDUSTRIAL":0.0,"RURAL":0.0,"VILLAGE":0.0},"segment_overrides":{}}

engineering_indicator.construction.rock_shield_present | 石方段保护层 | IMPORTANT

engineering_indicators.observations_by_segment.*.construction.rock_shield_present

值: {"by_archetype":{"DENSE":true,"GEOHAZARD":true,"INDUSTRIAL":true,"RURAL":true,"VILLAGE":true},"segment_overrides":{}}

engineering_indicator.construction.supervision_contractor | 监理单位 | IMPORTANT

engineering_indicators.observations_by_segment.*.construction.supervision_contractor

值: {"by_archetype":{"DENSE":"SYNTHETIC_SUPERVISION_CO","GEOHAZARD":"SYNTHETIC_SUPERVISION_CO","INDUSTRIAL":"SYNTHETIC_SUPERVISION_CO","RURAL":"SYNTHETIC_SUPERVISION_CO","VILLAGE":"SYNTHETIC_SUPERVISION_CO"},"segment_overrides":{}}

engineering_indicator.construction.weld_ndt_coverage_fraction | 焊缝无损检测覆盖率 | IMPORTANT

engineering_indicators.observations_by_segment.*.construction.weld_ndt_coverage_fraction

值: {"by_archetype":{"DENSE":1.0,"GEOHAZARD":1.0,"INDUSTRIAL":1.0,"RURAL":1.0,"VILLAGE":1.0},"segment_overrides":{}}

engineering_indicator.construction.weld_reject_rate_fraction | 焊口返修率 | IMPORTANT

engineering_indicators.observations_by_segment.*.construction.weld_reject_rate_fraction

值: {"by_archetype":{"DENSE":0.0,"GEOHAZARD":0.0,"INDUSTRIAL":0.0,"RURAL":0.0,"VILLAGE":0.0},"segment_overrides":{}}

engineering_indicator.construction.welding_process | 焊接工艺 | IMPORTANT

engineering_indicators.observations_by_segment.*.construction.welding_process

值: {"by_archetype":{"DENSE":"SAW_AUTOMATIC_COMBINED","GEOHAZARD":"SAW_AUTOMATIC_COMBINED","INDUSTRIAL":"SAW_AUTOMATIC_COMBINED","RURAL":"SAW_AUTOMATIC_COMBINED","VILLAGE":"SAW_AUTOMATIC_COMBINED"},"segment_overrides":{}}

engineering_indicator.management_emergency.alarm_response_time_s | 报警响应时间 | IMPORTANT

engineering_indicators.observations_by_segment.*.management_emergency.alarm_response_time_s

值: {"by_archetype":{"DENSE":0.0,"GEOHAZARD":0.0,"INDUSTRIAL":0.0,"RURAL":0.0,"VILLAGE":0.0},"segment_overrides":{}}

engineering_indicator.management_emergency.audit_nonconformity_open_count | 未关闭审核不符合项 | IMPORTANT

engineering_indicators.observations_by_segment.*.management_emergency.audit_nonconformity_open_count

值: {"by_archetype":{"DENSE":0,"GEOHAZARD":0,"INDUSTRIAL":0,"RURAL":0,"VILLAGE":0},"segment_overrides":{}}

现场检查记录扫描件

SYNTHETIC_TEST_ONLY | 合成测试数据 | 第 3 页

S40_MITIGATION_PACKAGE × D00_CLEAN | 基准日期 2026-08-26 | 禁止用于真实工程

engineering_indicator.management_emergency.emergency_drills_per_year | 年应急演练次数 | IMPORTANT

engineering_indicators.observations_by_segment.*.management_emergency.emergency_drills_per_year

值: {"by_archetype":{"DENSE":0.0,"GEOHAZARD":0.0,"INDUSTRIAL":0.0,"RURAL":0.0,"VILLAGE":0.0},"segment_overrides":{}}

engineering_indicator.management_emergency.emergency_isolation_procedure_available | 紧急隔离程序可用 | IMPORTANT

engineering_indicators.observations_by_segment.*.management_emergency.emergency_isolation_procedure_available

值: {"by_archetype":{"DENSE":true,"GEOHAZARD":true,"INDUSTRIAL":true,"RURAL":true,"VILLAGE":true},"segment_overrides":{}}

engineering_indicator.management_emergency.emergency_team_response_time_s | 应急队伍响应时间 | IMPORTANT

engineering_indicators.observations_by_segment.*.management_emergency.emergency_team_response_time_s

值: {"by_archetype":{"DENSE":0.0,"GEOHAZARD":0.0,"INDUSTRIAL":0.0,"RURAL":0.0,"VILLAGE":0.0},"segment_overrides":{}}

engineering_indicator.management_emergency.maintenance_completion_fraction | 维护计划完成率 | IMPORTANT

engineering_indicators.observations_by_segment.*.management_emergency.maintenance_completion_fraction

值: {"by_archetype":{"DENSE":0.0,"GEOHAZARD":0.0,"INDUSTRIAL":0.0,"RURAL":0.0,"VILLAGE":0.0},"segment_overrides":{}}

engineering_indicator.management_emergency.management_of_change_compliance_fraction | 变更管理符合率 | IMPORTANT

engineering_indicators.observations_by_segment.*.management_emergency.management_of_change_compliance_fraction

值: {"by_archetype":{"DENSE":0.0,"GEOHAZARD":0.0,"INDUSTRIAL":0.0,"RURAL":0.0,"VILLAGE":0.0},"segment_overrides":{}}

engineering_indicator.management_emergency.misoperation_event_count | 历史误操作事件数 | IMPORTANT

engineering_indicators.observations_by_segment.*.management_emergency.misoperation_event_count

值: {"by_archetype":{"DENSE":0,"GEOHAZARD":0,"INDUSTRIAL":0,"RURAL":0,"VILLAGE":0},"segment_overrides":{}}

engineering_indicator.management_emergency.operating_procedure_available | 运行规程完整 | IMPORTANT

engineering_indicators.observations_by_segment.*.management_emergency.operating_procedure_available

值: {"by_archetype":{"DENSE":true,"GEOHAZARD":true,"INDUSTRIAL":true,"RURAL":true,"VILLAGE":true},"segment_overrides":{}}

engineering_indicator.management_emergency.operator_competency_pass_fraction | 操作人员能力考核通过率 | IMPORTANT

engineering_indicators.observations_by_segment.*.management_emergency.operator_competency_pass_fraction

值: {"by_archetype":{"DENSE":0.0,"GEOHAZARD":0.0,"INDUSTRIAL":0.0,"RURAL":0.0,"VILLAGE":0.0},"segment_overrides":{}}

SYNTHETIC_TEST_ONLY | 合成测试数据 | 第 4 页

engineering_indicator.management_emergency.operator_training_completion_fraction | 操作人员培训完成率 | IMPORTANT
engineering_indicators.observations_by_segment.*.management_emergency.operator_training_completion_fraction
值: {"by_archetype":{"DENSE":0.0,"GEOHAZARD":0.0,"INDUSTRIAL":0.0,"RURAL":0.0,"VILLAGE":0.0},"segment_overrides":{}}

engineering_indicator.management_emergency.permit_to_work_compliance_fraction | 作业许可符合率 | IMPORTANT
engineering_indicators.observations_by_segment.*.management_emergency.permit_to_work_compliance_fraction
值: {"by_archetype":{"DENSE":0.0,"GEOHAZARD":0.0,"INDUSTRIAL":0.0,"RURAL":0.0,"VILLAGE":0.0},"segment_overrides":{}}

engineering_indicator.management_emergency.procedure_compliance_fraction | 规程执行符合率 | IMPORTANT
engineering_indicators.observations_by_segment.*.management_emergency.procedure_compliance_fraction
值: {"by_archetype":{"DENSE":0.0,"GEOHAZARD":0.0,"INDUSTRIAL":0.0,"RURAL":0.0,"VILLAGE":0.0},"segment_overrides":{}}

engineering_indicator.management_emergency.shift_handover_compliance_fraction | 交接班符合率 | IMPORTANT
engineering_indicators.observations_by_segment.*.management_emergency.shift_handover_compliance_fraction
值: {"by_archetype":{"DENSE":0.0,"GEOHAZARD":0.0,"INDUSTRIAL":0.0,"RURAL":0.0,"VILLAGE":0.0},"segment_overrides":{}}

engineering_indicators.observations_global | 全局工程指标观测 | IMPORTANT

engineering_indicators.observations_global

值: {'consequence_parameters.probit_model_version': {'as_of': '2026-08-04', 'quality': 'B', 'source_ref': 'AQ/T 3046-2013', 'value': 'AQT3046-2013-I.1'}, 'consequence_parameters.tno_curve_version': {'as_of': '2026-08-04', 'quality': 'C', 'source_ref': 'AQ/T 3046-2013 Figure E.1', 'value': 'AQT3046-FIG-E1-DIGITIZED-PROVISIONAL'}, 'governance.approval_ref': {'as_of': '2026-08-04', 'quality': 'D', 'source_ref': 'synthetic://approval', 'value': 'TEST-ONLY-NOT-APPROVED'}, 'governance.as_built_drawing_ref': {'as_of': '2026-08-04', 'quality': 'C', 'source_ref': 'synthetic://as-built', 'value': 'SYN-ASBUILT-001'}, 'governance.construction_recording_ref': {'as_of': '2026-08-04', 'quality': 'C', 'source_ref': 'synthetic://construction-recording', 'value': 'SYN-CONSTRUCTION-RECORDING-001'}}

engineering_indicator.governance.approval_ref | 模型与参数批准文件引用 | IMPORTANT
engineering_indicators.observations_global.governance.approval_ref
值: "TEST-ONLY-NOT-APPROVED"

```
engineering_indicator.governance.as_built_drawing_ref | 竣工图引用 | IMPORTANT
engineering_indicators.observations_global.governance.as_built_drawing_ref
值: "SYN-ASBUILT-001"
```

engineering_indicator.governance.construction_record_ref | 施工记录引用 | IMPORTANT
engineering_indicators.observations_global.governance.construction_record_ref
值: "SYN-CONSTRUCTION-001"

现场检查记录扫描件

SYNTHETIC_TEST_ONLY | 合成测试数据 | 第 5 页

S40_MITIGATION_PACKAGE × D00_CLEAN | 基准日期 2026-08-26 | 禁止用于真实工程

engineering_indicator.governance.data_owner | 数据责任人 | IMPORTANT

engineering_indicators.observations_global.governance.data_owner

值: "SYNTHETIC_DATA_OWNER"

engineering_indicator.governance.design_report_ref | 初步设计报告引用 | IMPORTANT

engineering_indicators.observations_global.governance.design_report_ref

值: "SYN-DESIGN-001"

engineering_indicator.governance.emergency_plan_ref | 应急预案引用 | IMPORTANT

engineering_indicators.observations_global.governance.emergency_plan_ref

值: "SYN-ERP-001"

engineering_indicator.governance.environment_report_ref | 环境影响评价报告引用 | IMPORTANT

engineering_indicators.observations_global.governance.environment_report_ref

值: "SYN-ENV-001"

engineering_indicator.governance.geohazard_report_ref | 地质灾害评估报告引用 | IMPORTANT

engineering_indicators.observations_global.governance.geohazard_report_ref

值: "SYN-GEO-001"

engineering_indicator.governance.hca_report_ref | 高后果区识别报告引用 | IMPORTANT

engineering_indicators.observations_global.governance.hca_report_ref

值: "SYN-HCA-001"

engineering_indicator.governance.integrity_plan_ref | 完整性管理计划引用 | IMPORTANT

engineering_indicators.observations_global.governance.integrity_plan_ref

值: "SYN-IMP-001"

engineering_indicator.governance.previous_risk_report_ref | 既往风险评价报告引用 | IMPORTANT

engineering_indicators.observations_global.governance.previous_risk_report_ref

值: "SYN-PREV-QRA-001"

现场检查记录扫描件

SYNTHETIC_TEST_ONLY | 合成测试数据 | 第 6 页

S40_MITIGATION_PACKAGE × D00_CLEAN | 基准日期 2026-08-26 | 禁止用于真实工程

engineering_indicator.governance.route_map_ref | 遥感或航拍线路图引用 | IMPORTANT

engineering_indicators.observations_global.governance.route_map_ref

值: "SYN-ROUTE-001"

engineering_indicator.governance.supervision_record_ref | 监理记录引用 | IMPORTANT

engineering_indicators.observations_global.governance.supervision_record_ref

值: "SYN-SUPERVISION-001"